

AVA POLAND

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EDUCATION

B.S. Mechanical Engineering, Minor in Physics | Kansas State University | Manhattan, KS | GPA: 3.4/4.0 | Expected Dec. 2028

- SolidWorks Certified — C-R7Z4VQB269

TECHNICAL SKILLS

Mechanical Design & CAD: SolidWorks, Onshape, Fusion 360, Autodesk Inventor, Autodesk Vault, Fusion Manage, Parametric Modeling, GD&T, Engineering Drawings, ANSI Drafting Standards

Manufacturing & Prototyping: FDM 3D Printing, CNC Machining, Laser Cutting, G-code, Design for Manufacturability (DFM), Manufacturing Systems, Prototyping

Programming & Automation: Python, MATLAB, iLogic, Parametric CAD Automation

ENGINEERING EXPERIENCE

Mechanical Engineering Intern | Automatic Systems Incorporated | Kansas City, MO | May 2026 – Aug. 2026

- Developed iLogic-based CAD automation templates that generated 50+ model and engineering drawing variants of conveyor components in minutes, replacing hours-to-days of repetitive manual modeling and drafting.
- Modeled and detailed mechanical components for power-and-free conveyor systems deployed in GM and Ford manufacturing facilities.
- Collaborated directly with shop technicians and manufacturing personnel to identify and resolve manufacturability issues while maintaining company design standards.
- Applied parametric CAD, engineering drawing standards, GD&T, and design-for-manufacturability principles to industrial mechanical systems.

Engineering Graphics Teaching Assistant, ME 212 | Kansas State University | Manhattan, KS | Jan. 2026 – Present

- Instruct students in mechanical drafting, CAD modeling, ANSI engineering drawing standards, and design communication.
- Demonstrate mechanical design techniques through hand-sketching and CAD-based modeling during laboratory instruction.
- Develop original CAD-based examination and laboratory content by modeling mechanical components designed to assess specific engineering graphics competencies.

ENGINEERING PROJECTS & ACTIVITIES

Pi Car Pro — Custom Four-Finger Robotic Gripper | Personal Engineering Project | 2026 – Present

- Designing a custom four-finger robotic gripper and articulated arm in Onshape for integration with a Pi Car Pro mobile robotic platform.
- Re-engineering a reference joint concept for reduced scale, practical assembly, and real-world manufacturability, adapting geometry originally developed as a non-production CAD concept.
- Developing a hybrid manufacturing strategy using TPU for compliant gripping surfaces and machined aluminum for higher-load structural components.
- Designing the mechanism around available additive and subtractive manufacturing processes, with emphasis on assembly, mechanical integration, and component manufacturability.
- Developing the project's electronic and actuation architecture independently, including future motor/servo selection, controls, and system integration.

Wildcat Combat Robotics — Mechanical Design & Build Team | Kansas State University | Aug. 2026 – Present

- Contributing to mechanical design and CAD development of a competitive eggbeater-style combat robot, including system layout, component geometry, and iterative mechanical design.
- Supporting mechanical/electrical integration and hardware testing across drivetrain and robot subsystems.
- Evaluating mechanical constraints and iterating components to improve functionality, manufacturability, and system integration.

FTC Tungsten Tigers — Build Team Captain | Blue Valley High School | Sept. 2022 – May 2024

- Co-led the mechanical design and fabrication of a multi-award-winning FTC competition robot, including machining chassis components and fabricating competition mechanisms.
- Designed and fabricated a robotic claw mechanism and designed and 3D-printed a paper-airplane launcher for competition use.
- Mentored team members in CAD, fabrication, prototyping, and robot construction.

AWARDS & HONORS

- 2nd Place — Innovate Award, FTC Kansas & Missouri State Competition | 2024
- 1st Place — Inspire Award, FTC Kansas City West League | 2024
- 4th Place — Engineering Manufacturing CAD Competition, Technology Student Association | 2024
- 1st Place — Design Award, FTC Kansas City West League | 2023